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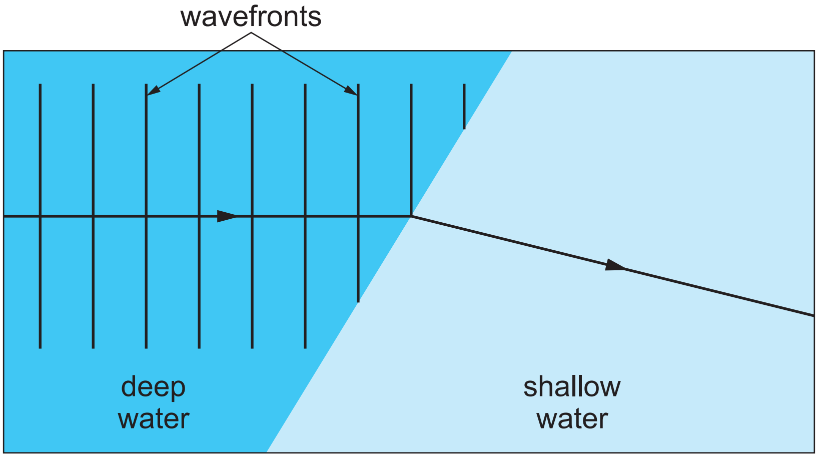
2. Water waves refract when they travel from deep water to shallow water.

Refraction happens because the speed of the wave changes.

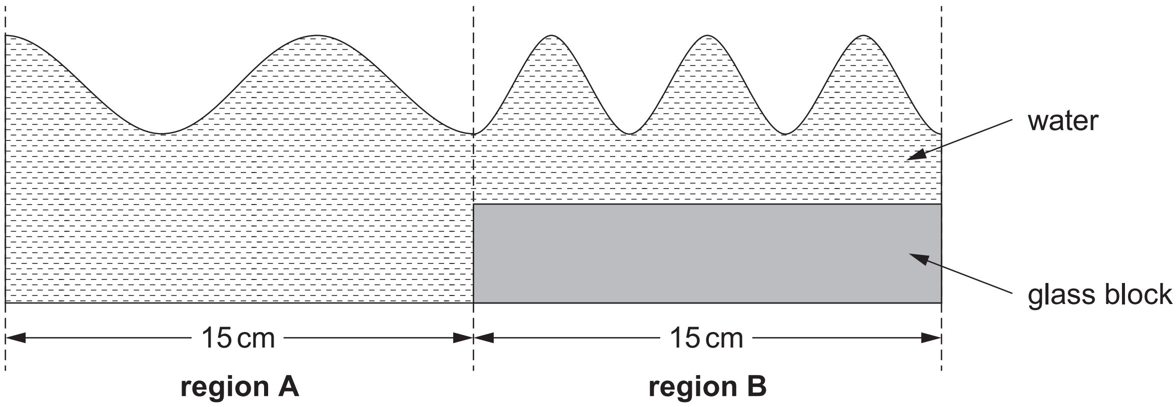
The wavelength also changes because the frequency is constant.

Students use a ripple tank to investigate this effect.

(a) **Complete the diagram below** to show the water waves in shallow water. [3]



(b) The depth of the shallow water can be changed by using glass blocks of different thicknesses.



Regions **A** and **B** are both 15 cm long.

(i) I. How many waves are shown in **region A**? .... [1]

II. Calculate the wavelength of the waves in **region A**. [1]

wavelength = ... cm

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(ii) John says that the wave speed in **region B** is greater than the wave speed in **region A**. Explain whether John is correct. [2]

....

....

....

(c) In another experiment using a different tank, students investigate how the depth of water affects wave speed.

They change the depth of the water using different thickness glass blocks.

The water level is kept constant at 10 cm.

The table below shows their results.

Thickness of glass block (cm)	Depth of water (cm)	Wave speed (cm/s)
8	2	60
6	4	75
4	....	82

(i) **Complete the table.** [1]

(ii) Use the equation:

$$\text{wavelength} = \frac{\text{wave speed}}{\text{frequency}}$$

to calculate the wavelength of water waves of frequency 50 Hz when the thickness of the glass block is 6 cm. [2]

wavelength = ... cm

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(iii) Janet states that when the thickness of the glass block decreases by 2 cm the wave speed increases by a quarter.
Explain to what extent Janet is correct. [3]
Space for calculations.

....

....

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